

263-4600-00L Formal Methods for Information Security

Semester	Spring Semester 2024
Lecturers	S. Krstic, R. Sasse, C. Sprenger
Periodicity	yearly recurring course
Language of instruction	English

◀ Tabs

Courses

Number	Title	Hours	Lecturers
263-4600-00 V	Formal Methods for Information Security	2 hrs Tue 10:15-12:00 CAB G 59 »	S. Krstic, R. Sasse, C. Sprenger
263-4600-00 U	Formal Methods for Information Security	1 hrs Tue 12:15-13:00 CAB G 59 »	S. Krstic, R. Sasse, C. Sprenger
263-4600-00 A	Formal Methods for Information Security	1 hrs	S. Krstic, R. Sasse, C. Sprenger

Catalogue data

Abstract	The course focuses on formal methods for the modeling and analysis of security protocols for critical systems, ranging from authentication protocols for network security to electronic voting protocols and online banking. In addition, we will also introduce the notions of non-interference and runtime monitoring.
Learning objective	The students will learn the key ideas and theoretical foundations of formal modeling and analysis of security protocols. The students will complement their theoretical knowledge by solving practical exercises, completing a small project, and using state-of-the-art tools. The students also learn the fundamentals of non-interference and runtime monitoring.
Content	The course treats formal methods mainly for the modeling and analysis of security protocols. Cryptographic protocols (such as SSL/TLS, SSH, Kerberos, SAML single-sign on, and IPSec) form the basis for secure communication and business processes. Numerous attacks on published protocols show that the design of cryptographic protocols is extremely error-prone. A rigorous analysis of these protocols is therefore indispensable, and manual analysis is insufficient. The lectures cover the theoretical basis for the (tool-supported) formal modeling and analysis of such protocols. Specifically, we discuss their operational semantics, the formalization of security properties, and techniques and algorithms for their verification. The second part of this course will cover a selection of advanced topics in security protocols such as abstraction techniques for efficient verification, secure communication with humans, the link between symbolic protocol models and cryptographic models as well as RFID protocols (a staple of the Internet of Things) and electronic voting protocols, including the relevant privacy properties.

Moreover, we will give an introduction to two additional topics: non-interference as a general notion of secure systems, both from a semantic and a programming language perspective (type system), and runtime verification/monitoring to detect violations of security policies expressed as trace properties.

Competencies 	Subject-specific Competencies	Concepts and Theories	assessed
	Method-specific Competencies	Analytical Competencies	assessed
		Problem-solving	assessed
		Project Management	fostered
	Social Competencies	Cooperation and Teamwork	fostered
	Personal Competencies	Self-direction and Self-management	fostered

Performance assessment

Performance assessment information (valid until the course unit is held again)	
▶ Performance assessment as a semester course	
ECTS credits	5 credits
Examiners	C. Sprenger , S. Krstic, R. Sasse
Type	session examination
Language of examination	English
Repetition	The performance assessment is offered every session. Repetition possible without re-enrolling for the course unit.
Mode of examination	oral 25 minutes
Additional information on mode of examination	The grade is determined by a project [25%] and the final oral exam [75%]. The compulsory project will be worked on by teams of two students.
This information can be updated until the beginning of the semester; information on the examination timetable is binding.	

Learning materials

Main link	Formal Methods for Information Security
Only public learning materials are listed.	

Groups

No information on groups available.

Restrictions

There are no additional restrictions for the registration.

Offered in

Programme	Section	Type	
CAS in Computer Science	Focus Courses and Electives	W	
Cyber Security Master	Electives	W	
DAS in Cyber Security	Elective Courses	W	
Computer Science Master	Elective Courses	W	

Computer Science Master	Minor in Information Security	W	
Computer Science Master	Minor in Programming Languages and Software Engineering	W	